Adroit Technologies Protocol Driver

Name	ScanRS Barcode Serial Scanner
DLL	SCANRS

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Contents

1. Overview

Adroit supports communication with the following devices:

- Scanners that return a string (unsolicited data, appended with carriage return (**<CR>**) and line feed (**<LF>**) control characters at the end)
- Any device that returns an ASCII string followed by a common terminating sequence of some kind. For example, <CR> and <LF>.

Gryphon_D100 or UNRESTRICTED, for the other Unknown models. This driver has been tested with a Gryphon_D100 on site and only supports unsolicited reception of a data string used for monitoring barcodes.

PLEASE NOTE

Only string type Adroit items may be scanned using this driver (e.g., Text, String, etc) with exception for the Administrative register which is only scannable to Integer type slots. If data is required in Integer, Reals or Booleans slot types, then try using the Adroit "USERDRV" Driver which is script configurable.

PLEASE NOTE

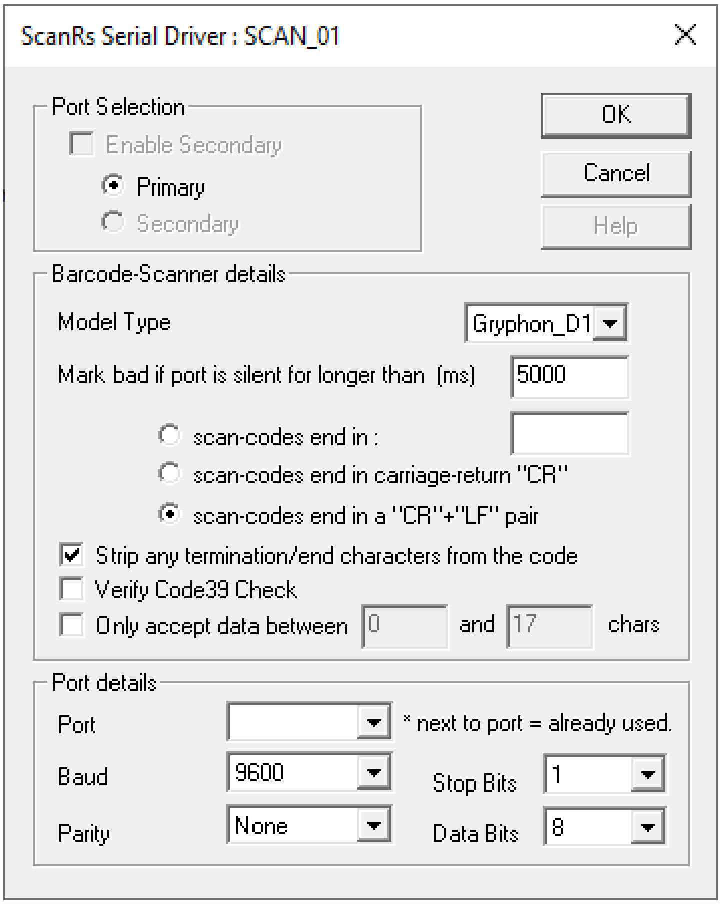
Each device is treated as a point-to-point device i.e. One COM port per device. Attempting to scan more than one device will result in driver device attachment failure.

2. Wiring

Not applicable.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection (Not available)

Not implemented.

3.1.2. Barcode Scanner Details

Model Type - (Default = Gryphon-D100) Select 'Unrestricted' if the device is not listed.

Mark bad if port is silent for longer than (mSec) - (Default = 5000) Insert the required deadband to instruct the driver to delay before marking all Adroit scanned items bad if the port has not received any data. Setting it to 0 will disable marking bad.

Scan Codes end in : - Select this if a non-standard termination sequence is required. Type any number of characters (Printable characters only).

Scan Codes end in "CR" - Select this if the device ends its string sequence with a "CR" (carriage return control character).

Scan Codes end in "CR" & "LF" – (Default) Select this if the device ends its string sequence with <CR> and <LF>.

Strip any termination/end characters from the code - (Default = checked)

- If CHECKED: the driver will search for these characters to determine the message length, thus buffering the incoming data until a CR and LF is encountered. Failing this, the buffer may overflow and an appropriate error message will be generated.

- If UNCHECKED: upon any data received, the driver will attempt to unpack into all scanned jobs even if only partially filling the range. It is up to the user to check via script to validate the data received.

Verify code39 Check – (Default = unchecked) Check this checkbox if the device is using code39-modulo43 check-summation.

Only Accept data between – (Default = unchecked) Check this checkbox to force the driver to throw away any full reads (i.e. when a valid string terminator is found) that are smaller or larger than the bytes specified. Min/Max bytes is inclusive and includes terminating characters.

3.1.3. Port Details

Port - The available serial ports on the machine.

PLEASE NOTE
If a * next to a port is seen, it indicates the port is already in use by Adroit, either by this driver as another device, or another application such as a modem/Configurator etc.

Baud – (Default = 9600) (...,9600, 19200, 38400, 57600, ...)

Parity – (Default = None)

Stop Bits – (Default = 1)

Data Bits – (Default = 8)

3.2. Supported Model Ranges

- "Gryphon_D100" - DB Offset range 0 – 255 bytes.
- "Gryphon_D100" - A Offset range 0 register.
- "Unrestricted" - DB Offset range 0 – 8191 bytes.
- "Unrestricted" - A Offset range 0 register.

Address types and maximum ranges supported by the Adroit ScanRS Protocol driver.

4. Supported Addresses

Address types and maximum ranges supported by the Adroit ScanRS Protocol driver.

Device	Data Type	Example	Integer	Boolean	Real	String
Data Byte Area	String	DB:0 S20	-	-	-	Yes
Admin Area	Integer	A:0	Yes	-	-	-

PLEASE NOTE

The size setup in the device configuration, "only accept between __ and __ chars" in the dialog is tested before any unpacking to any strings in the driver. If a violation is found, no unpacking will take place and the read is considered invalid and will be rejected.

4.1. Administration Area/Registers

Addressing structure:

[**Admin Type**]:[**Offset**]

Type is 'A' Admin register.
Offset can only be 0 at this time. Note: zero based indexing is used.

Example address:

A:0

Scanning the address A:0 to a Marshal Agent or any Integer slot will allow the user to get additional information regarding the last read transaction. For more information, see the following table. These can be alarmed or used for triggers, etc.

PLEASE NOTE

Digitals are NOT supported. Please use Marshal Agents and access those bits individually using the mechanism provided by the Agent.

Device	Data Type
Bit 0	The last read was out of range based on size rule configured in setup.
Bit 1	CRC failure.
Bit 2-15	Not used

4.2. Data Byte Area/Registers

Address structure:

[**Data byte Type**]:[**Offset**] **S:**[**Maximum String Size**]

Type is 'DB' Data byte.
Offset is any valid offset within the expected receive range (0 to 255), zero based indexing is used.
String Size – Any length but not bigger than 80, and not 0 or lower.

Example:

The barcode:

123xy456abc-01<CR><LF>

Scanning address DB:0 S20 to a String Agent will result in **123xy456abc-01** in the value slot. (Spaces 15-20 in the String Agent are not used.)

Scanning address DB:0 S4 to another String Agent will result in **123x** appearing in the value slot.

Scanning address DB:9 S1 to another string agent will result in **b** appearing in the value slot.

Scanning address DB:7 S20 to another string agent will result in **6abc-01** appearing in the value slot. Even though a size of 20 was specified, only the remaining 7 characters were available, and thus inserted. This is not considered an error because the terminating sequence was found.
These four agents overlap but that does not matter. All these string agents are filled in after every successful read. Tags may overlap only if offset sizes differ else a duplicate is assumed and handled accordingly.

These examples assume terminating control characters **<CR><LF>** were set to be stripped (according to the settings in the device configuration.)

5. Protocol Definition and Considerations

5.1. Protocol Overview

Any ASCII data + Terminating sequence.
The Terminating sequence MUST be specified else the read will not work.
E.g. DATA + (CR+LF) as terminating bytes.

6. General Notes and Troubleshooting

6.1. Contact Details

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Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	17-Jan-2023	K. Robinson	Grammar, spelling edits, alignment, and content editing.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
5	27/07/2002	1st Release of ScanRS string driver.
10	5/11/2002	Modulo43 checkSummation added.
11	27/11/2002	Admin reg added, and bytes range check added.
12	6/01/2003	Doc refined, Admin reg bit 1 added.